

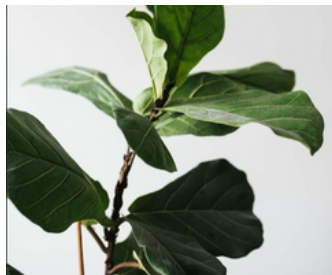
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AN-NAJAH NATIONAL UNIVERSITY



**Faculty of Engineering & Information Technology
Computer Engineering Department
Graduation Project I (Software)**

Plantsia



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**Presented in partial fulfillment of the requirements for Bachelor
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Yours sincerely,

Besan and Tasneem

Disclaimer

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Abstract

Plantsia is an innovative software project designed as an e-commerce platform specialized in selling plants and agricultural supplies. It was developed using modern technologies including Flutter for the frontend, Node.js for the backend, and MongoDB as the database. The system serves three main user roles: Admin, Shop Owner, and User, each with distinct responsibilities that ensure smooth operation and management of the platform.

The Admin oversees the platform by managing user roles (adding, editing, or deleting), viewing sales analytics, and adding new shop owners to expand the online marketplace. Shop Owners manage their shops by adding and updating plant and supply listings, receiving and processing orders, and handling modifications and deletions as needed. Users can browse and order plants and supplies, manage their favorite plants list, and use AI-based image analysis to diagnose plant diseases, understand causes, and receive treatment suggestions. Additionally, the platform provides a powerful search function for enhanced user convenience.

Plantsia demonstrates how software technologies can modernize agricultural commerce, promote digital transformation in the agricultural sector, and provide intelligent solutions to meet evolving market needs while supporting sustainability and innovation in smart farming.

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Chapter 1: Introduction

1.1 Statement of the Problem

The online plant market faces challenges due to the lack of integrated systems that support flexible shop management, accurate sales tracking, and smart features like advanced search and plant disease diagnosis. This calls for an innovative software solution that simplifies operations and enhances the user experience.

1.2 Objectives

The primary objective of this project is to design and implement an integrated e-commerce platform for selling plants and agricultural supplies. Specific goals include:

- Providing flexible shop and product management for admins and shop owners.
- Enabling users to browse, search, and order plants and supplies easily.
- Integrating smart features such as AI-based plant disease diagnosis and sales analytics.
- Supporting remote management and a user-friendly experience through modern technologies.
-

1.3 Scope of the Work

The project focuses on designing and developing an integrated software platform for selling plants and agricultural supplies. It covers the implementation of user roles (Admin, Shop Owner, User), shop and product management, order processing, AI-based plant disease diagnosis, and sales analytics. The scope does not include large-scale commercial deployment, marketing, or hardware development.

1.4 Significance of the Work

This platform addresses the demand for an efficient, flexible, and intelligent solution for buying and managing plants and agricultural supplies. By simplifying operations for shop owners and enhancing the user experience with smart features, the system provides a cost-effective and modern alternative that bridges the gap between traditional plant commerce and digital transformation.

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1.5 Organization of the report

This report clearly explains the design and implementation of the Plantsia project. It begins with the problem statement and background, followed by a detailed methodology covering system architecture, user roles, and software technologies used. The results section highlights the key functionalities achieved and testing outcomes, while the discussion analyzes challenges faced and possible improvements. Finally, the conclusion summarizes the main findings and suggests directions for future development and enhancements.

Chapter 2: Literature Review

Plants and agricultural supplies are essential components of daily life, with plants playing a significant role in home decoration, environmental sustainability, and food production. The trade of plants has evolved over time from local markets and nurseries to modern e-commerce platforms that cater to a growing demand for convenience, variety, and smart features.

The process of managing plant sales involves several key stages, including shop setup, product listing, order processing, and customer service. Effective management of these stages is crucial for ensuring customer satisfaction, supporting shop owners, and maintaining high service quality. Traditionally, plant sales have relied on manual processes that often result in inefficiency, limited scalability, and inconsistent customer experiences.

Recent advancements in software technologies have transformed plant commerce. E-commerce platforms, AI-based plant disease diagnostics, and integrated sales analytics have enabled shop owners and customers to interact more efficiently. Modern solutions now provide features such as flexible shop management, personalized product recommendations, and remote access to platform functionalities, reflecting the consumer's growing preference for smart and customized services.

Given the increasing demand for plants and agricultural supplies, this project aims to design and implement Plantsia—an innovative software platform that simplifies and enhances plant commerce. The system allows admins to manage user roles and monitor sales performance, shop owners to manage their shops and orders, and users to browse, order, and benefit from AI-powered plant disease diagnosis. The platform integrates advanced technologies such as Flutter for the frontend, Node.js for the backend, and MongoDB for data management, while offering smart features like sales analytics and AI image analysis to meet modern market needs.

By combining software technologies and intelligent automation, Plantsia addresses the challenges of traditional plant commerce and delivers a flexible, user-friendly, and efficient solution that aligns with the evolving expectations of today's consumers.

Chapter 3: Constraints, Standards/ Codes and Earlier course work

3.1 Constraints:

During the execution of the Plantsia project, several constraints were encountered:

- **Technical Challenges:** Integrating advanced features such as AI-based plant disease diagnosis and sales analytics required extensive research and testing, which increased development time.
- **Internet Connectivity:** Occasional instability in internet connectivity during testing phases affected the ability to work efficiently on remote operations and data synchronization features.
- **Time Management:** Balancing academic commitments alongside the demands of designing and implementing a full-stack platform posed a significant time management challenge.

3.2 Standards/Codes

The following standards and codes were considered in the Plantsia project:

- **Security Standards:** Ensured compliance with best practices in data security and user privacy, including proper handling of authentication tokens and encrypted connections (e.g., HTTPS).
- **API Standards:** Followed RESTful API design principles for communication between the frontend and backend systems to ensure scalability and maintainability.
- **Database Standards:** Applied standard practices for data integrity and consistency using MongoDB, including validation rules and indexing for performance optimization.
- **Wireless Standards:** Utilized IEEE 802.11 for enabling wireless connectivity where applicable, particularly in remote or mobile access features.

3.3 Earlier Coursework

This project benefited significantly from knowledge gained in previous coursework:

- **Software Engineering and Web Programming:** Provided us with essential skills in designing, developing, and maintaining full-stack web applications, which were fundamental in building the Plantsia platform.
- **Database Systems:** Equipped us with knowledge on designing and managing databases, ensuring data integrity and optimizing queries, which was crucial for implementing Plantsia's data storage and retrieval systems.
- **Artificial Intelligence:** Introduced us to the basics of AI concepts and applications, which supported the integration of AI-based plant disease diagnosis in the platform.
- **Computer Networks:** Gave us a solid understanding of networking principles, enabling us to implement secure communication between the frontend and backend and support features such as remote access.
- **Research skills and Critical thinking:** Enhanced our ability to write structured reports, document system design, and rely on credible references, which contributed to preparing professional documentation for the project.

Chapter 4: Methodology

This chapter presents the methodology adopted in developing the Plantsia platform. It covers the system architecture, software components, overall design, and the integration of the mobile application and backend services.

4.1 Choosing the idea

The initial and most critical step in the project was selecting the idea. As a team passionate about plants and their importance in our daily lives and environment, we wanted to contribute to this field by combining our love for plants with technology. While exploring different ideas, we realized the challenges in traditional plant commerce, such as inefficiency, limited accessibility, and the absence of smart solutions. This inspired us to develop Plantsia, an innovative platform aimed at enhancing the online sale of plants and agricultural supplies. Our goal was to create a system that not only simplifies management for shop owners but also offers an enjoyable, flexible, and modern shopping experience for plant lovers.

4.2 Architecture & Technologies Utilized

To create an engaging, efficient, and scalable platform for plant commerce, we carefully selected modern tools and a robust architecture that ensures seamless operation for both mobile and web users. Here's an overview of how the system components come together:

4.2.1 Frontend Design

The frontend focuses on delivering a visually appealing and user-friendly interface that works smoothly on smartphones, tablets, and computers.

- Why Flutter?
- Flutter was our top choice as it enables us to build a single application that runs seamlessly on both mobile devices and web browsers. Its rich widget library simplifies creating responsive layouts and smooth animations.
- Key Features:
 - A consistent, attractive user interface across all devices.
 - Simple navigation for browsing plants, supplies, orders, and profiles.
 - Fluid animations and smooth transitions for a pleasant shopping experience.

4.2.2 Backend Operations

The backend manages all critical operations behind the scenes, including user authentication, order processing, plant listings, and AI-based diagnosis.

- Why Node.js and MongoDB?
- Node.js: Ensures high performance and fast response times, even with many concurrent users.
- MongoDB: Offers a flexible, schema-less structure ideal for managing plant data, user profiles, orders, and AI diagnostic records.
- Key Features:
 - Secure login and role management for admins, shop owners, and users.
 - Efficient handling of product management, order submissions, and favorites.
 - Real-time updates for order status and stock levels.

4.2.2 Backend Operations

One of the core innovations of Plantsia is the integration of artificial intelligence to enhance user experience and provide smart plant care solutions.

- Why AI?
- AI was integrated to empower the platform with intelligent features such as plant disease diagnosis, personalized recommendations, and data-driven insights. This helps users make informed decisions about plant care and supports shop owners in offering better services.
- Key Features:
 - Plant Disease Diagnosis: Allows users to upload images of plants and get instant feedback on potential diseases, causes, and suggested treatments.
 - Data-Driven Recommendations: Provides personalized suggestions for plants and agricultural supplies based on user preferences and interaction history.
 - Continuous Learning: The AI model can be trained and improved over time by incorporating new plant data and feedback to enhance accuracy.

4.3 Features & Implementation

The application provides many features and a lot to talk about. This section will focus on the application features and the role of each library in the implementation.

- **login &signup**

The application provides a secure and structured login and signup system tailored for different user roles: User, Shop Owner, and Admin

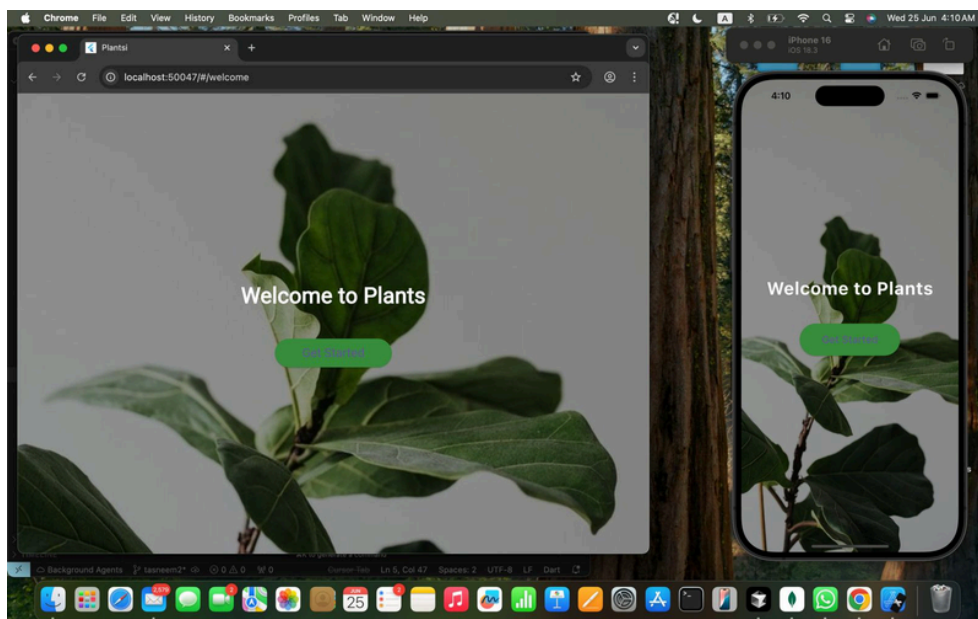


Figure 4.1: Welcome page.

- **User Signup & Login:**

When a user opens the app for the first time, they are prompted to sign up by providing their basic information (name, email, password, contact details). After creating the account, the user can log in anytime to browse plants, place orders, manage favorites, and access AI-powered plant diagnosis and recommendations.

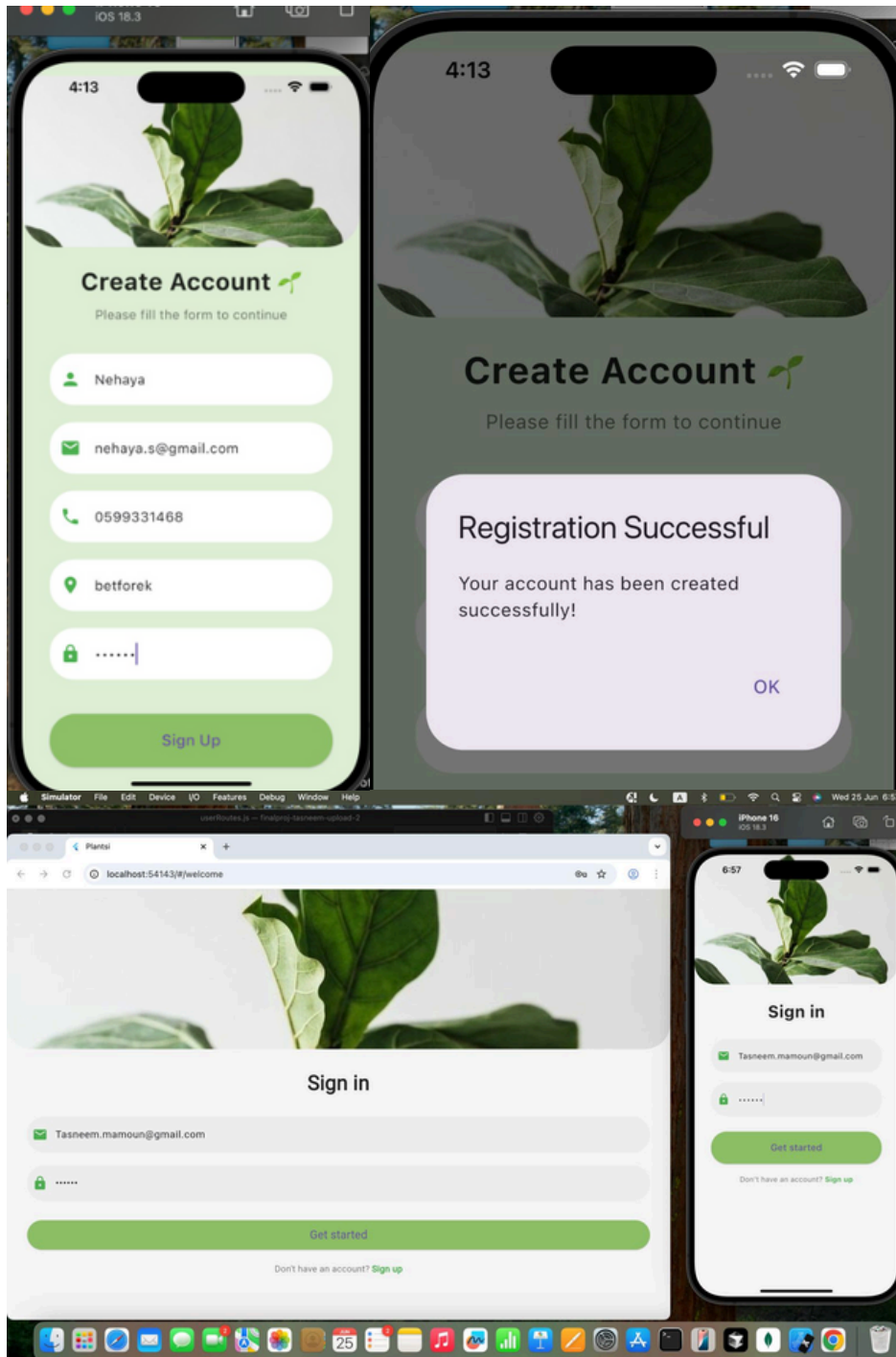


Figure 4.2: user login and signup.

• Shop Owner Account Creation & Login

Shop owners do not create their accounts on their own. Instead, the Admin is responsible for creating the shop owner's account within the admin dashboard. Once the admin creates the account, the shop owner can log in using the provided credentials and start managing their shop, products, and orders.

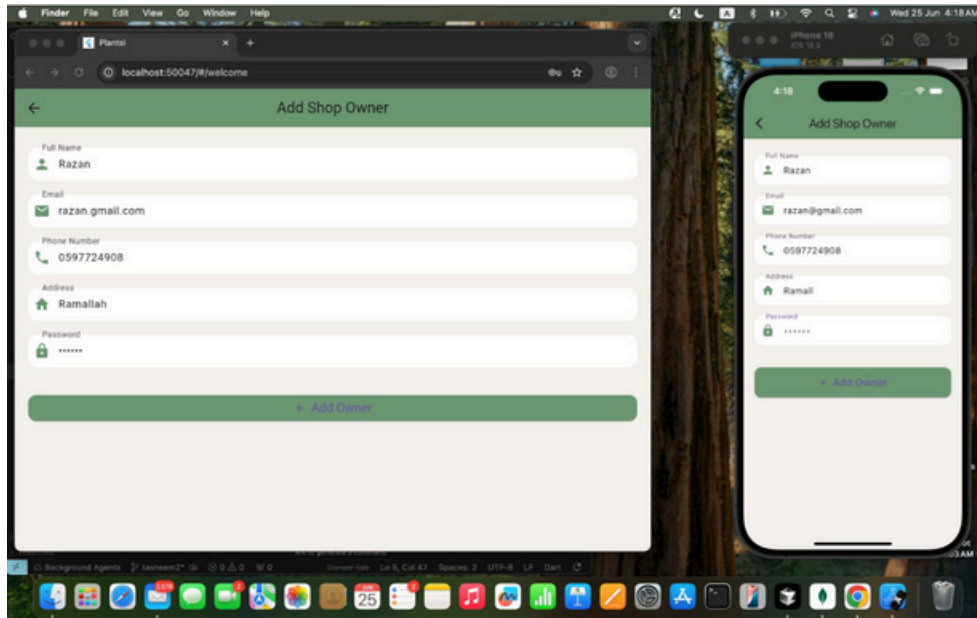
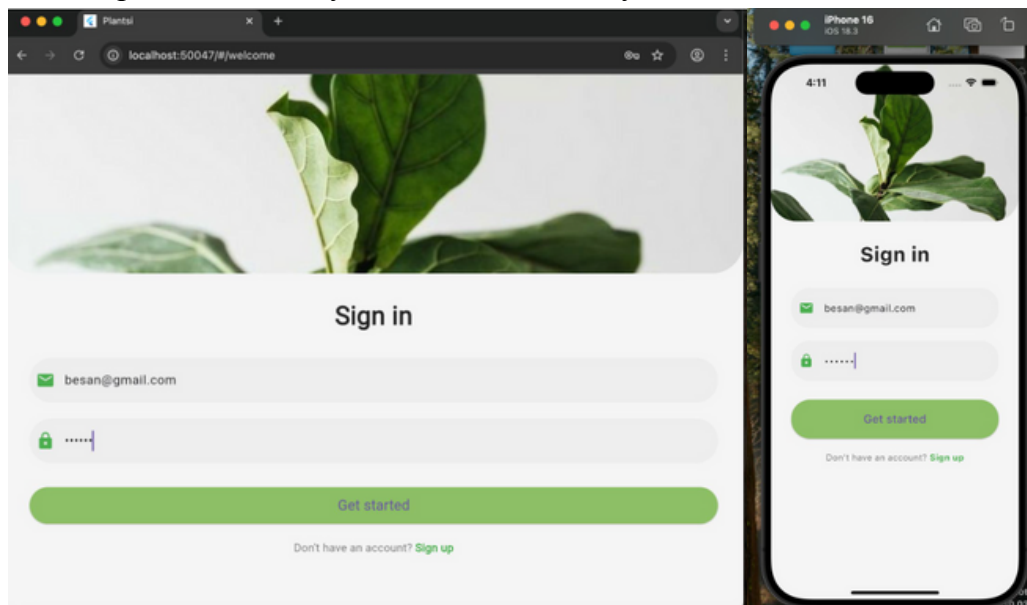


Figure 4.3: Shop Owner Account Creation & Login.

• Admin Login

Admin accounts are predefined during system setup. The admin does not need to sign up through the app. The admin simply logs in directly to manage user roles, add shop owners, oversee platform activity, and view sales analytics.



```
_id: ObjectId('6829136a71ea53d3e2c149b4')
name: "besan"
email: "besan@gmail.com"
password: "$2b$12$tQpw/IRKdu0S8y4UeCEHz.x1F0uoSShtia7Wbs20umVx0YFsSnkvC"
role: "admin"
__v: 0
```

Figure 4.4: Admin Login.

- **Admin Dashboard Overview**

This image shows the Admin Dashboard of the Plantsia application. Here, we've logged in as an admin, who holds the highest level of access within the system. The dashboard is designed to be clean, intuitive, and functional, allowing the admin to manage the platform effectively.

- The admin is granted full authority to:
- Add new Shop Owners.
- View and manage all registered Users.
- View all existing Shops on the platform.
- Track and manage Shop Owners.
- Monitor the Sales Ratio to analyze store performance.
- Log out of the system when done.
- In the following sections, we will explain each of these features in detail, with supporting images demonstrating how the admin interacts with each functionality.

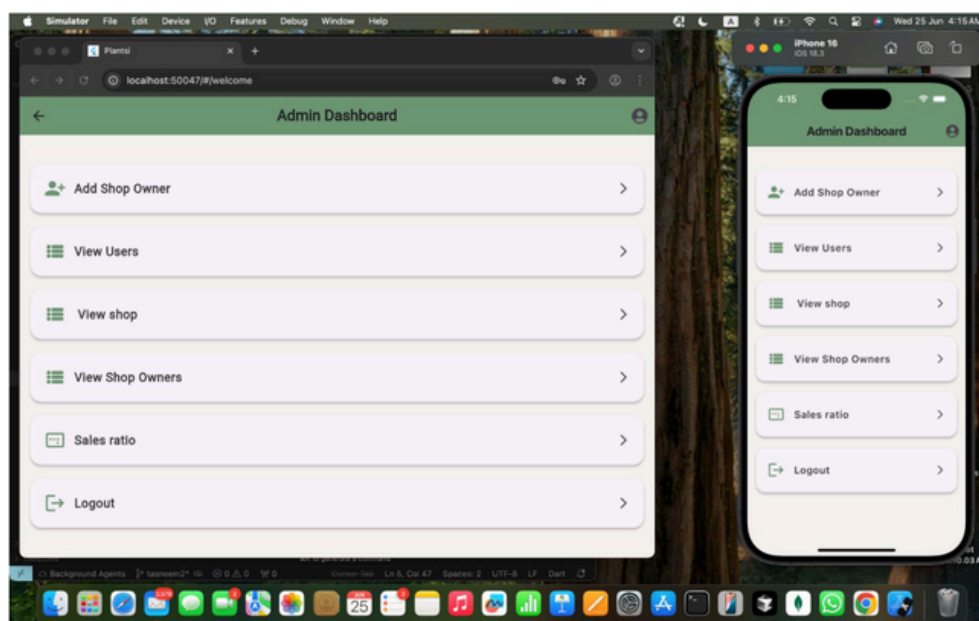


Figure 4.5: Admin Dashboard Overview.

1. Add Shop Owner

- This feature allows the admin to create a new Shop Owner account. Since shop owners cannot sign up directly through the app, the responsibility of creating their accounts falls entirely on the admin. The admin fills in the required information (such as name, email, and password), and the system stores the credentials securely.
- Once created, the shop owner can then log in using the provided credentials and begin managing their shop, products, and orders.
- This functionality ensures that only verified and approved individuals can manage shops, maintaining the platform's security and professionalism.

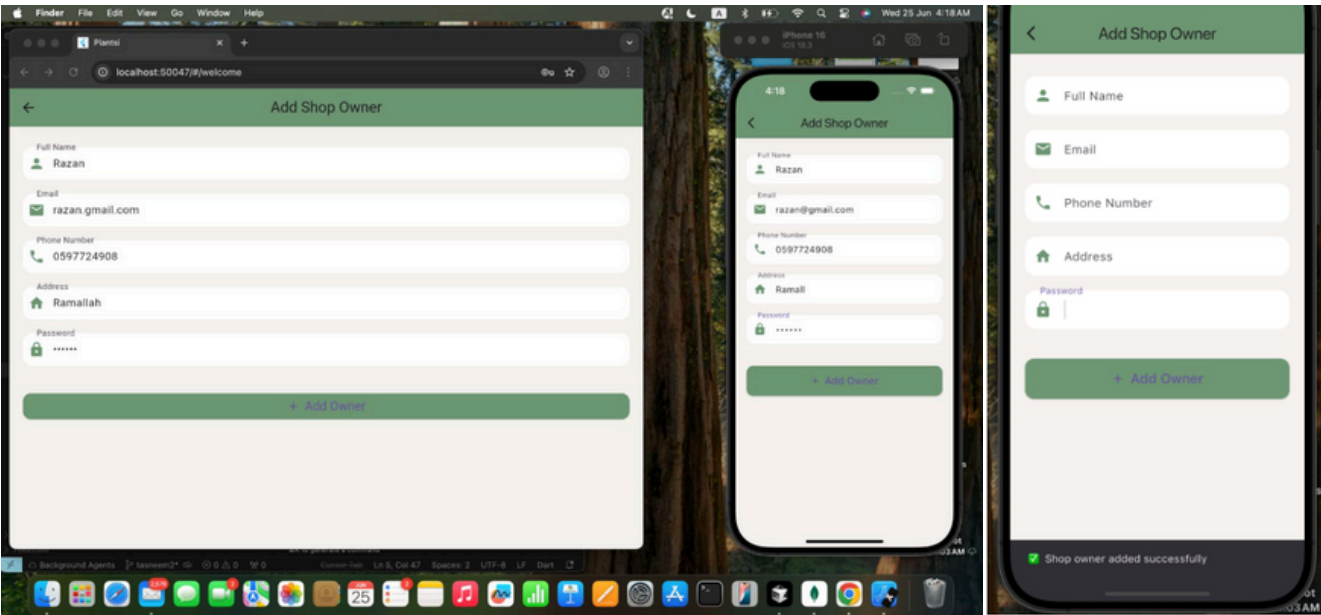


Figure 4.6: Add Shop Owner.

2.View Users

- Through this page, the admin can view all registered user accounts in the application. Details such as the user's name, email.

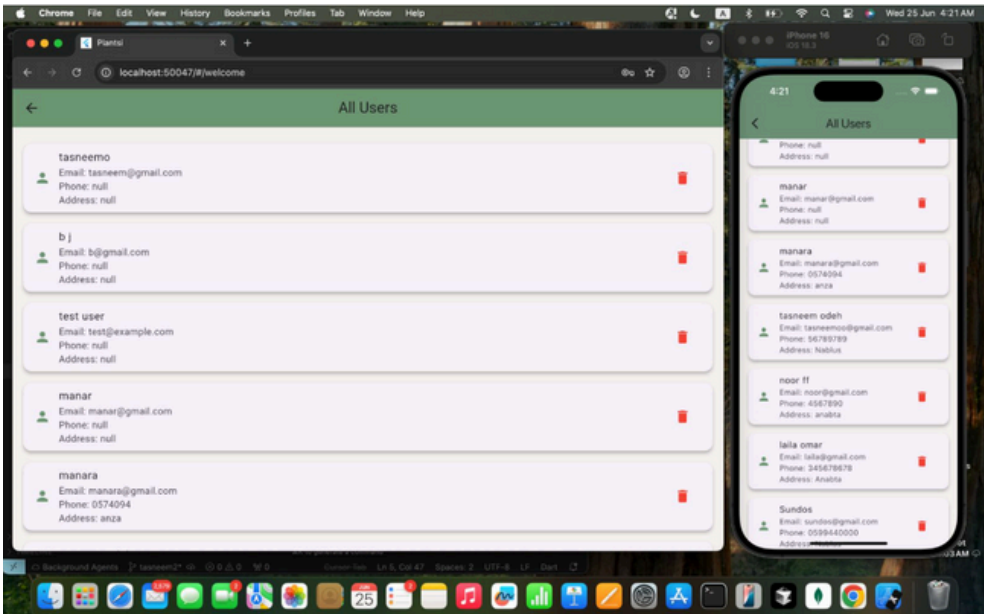


Figure 4.7: View Users.

3.View shops

- This feature allows the admin to see all the shops that have been added to the system by shop owners.

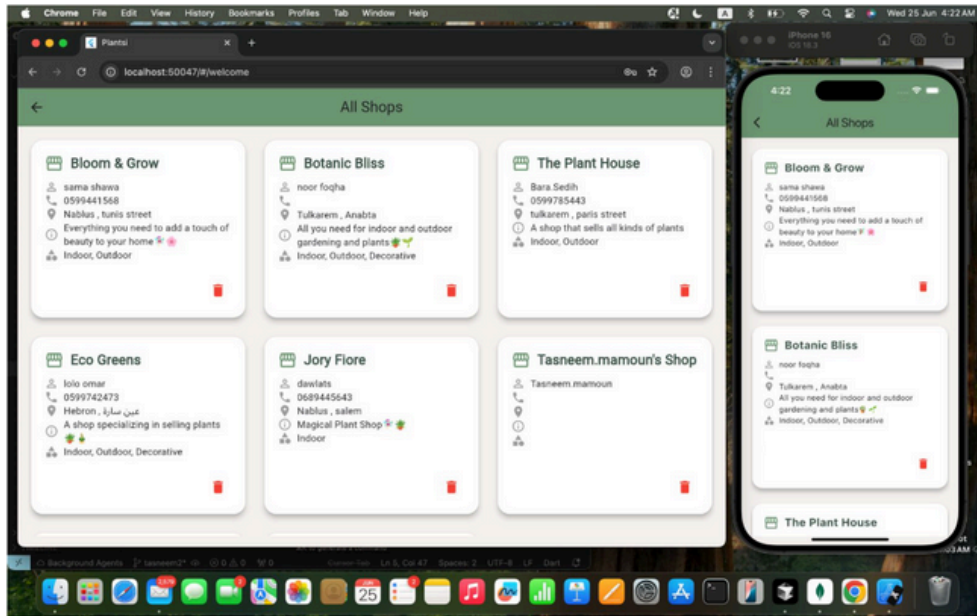


Figure 4.8: View Shops.

4.View shop owners

- Here, the admin can view a list of all the shop owners added to the system. .

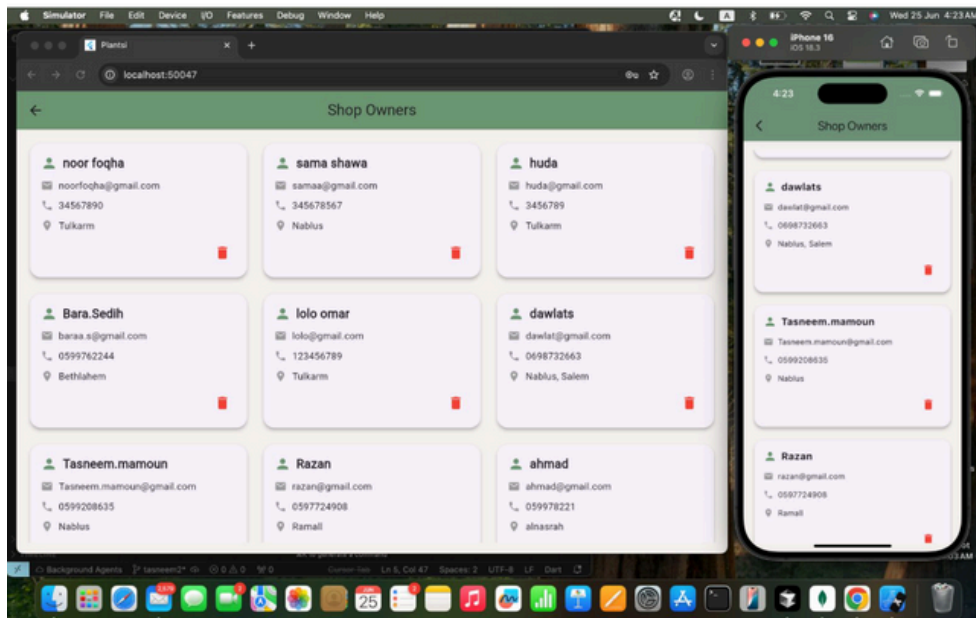


Figure 4.9: View Shop Owners.

5.Sales Ratio

This page displays detailed sales analytics for the entire platform. It allows the admin to track how each shop is performing in terms of sales.

Key insights shown in this section include:

- The total number of sales for each shop.
- A ranking of shops based on their sales volume, highlighting the top-performing stores.
- The most sold item in each shop, giving a clear idea of customer preferences.

This feature is useful for monitoring overall business performance, identifying successful products, and supporting strategic decisions like promotions or highlighting trending items.

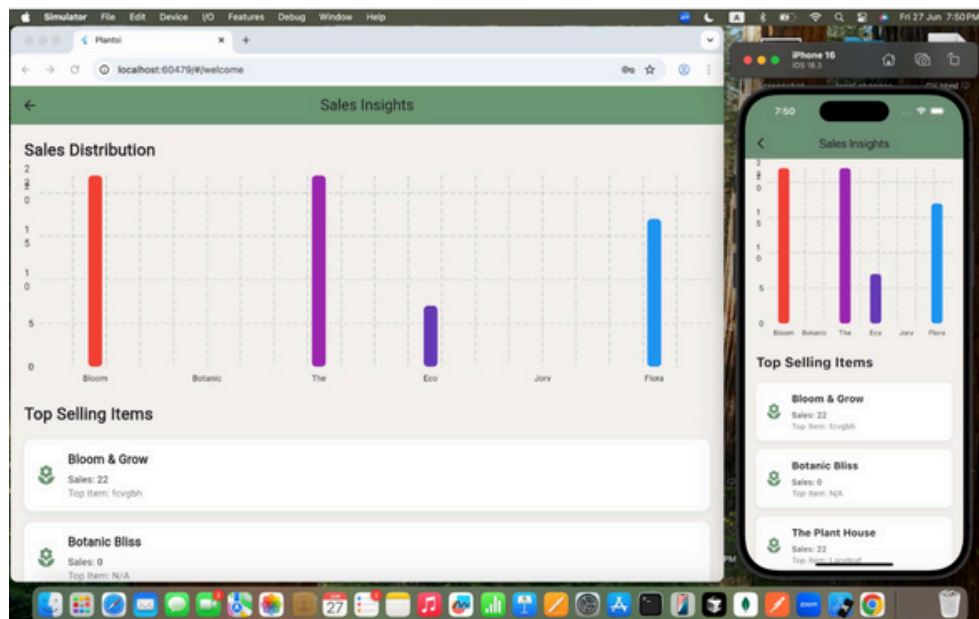


Figure 4.10: Sales Ratio.

• Profile Page

This page allows users to view and manage their personal information within the online plant store. Upon logging in, users are greeted by name and can see their profile photo along with essential account details.

Main features include:

- Edit Info:
 - Users can update their name, email address, phone number, and physical address to keep their profile up to date.
- Profile Picture:
 - Users have the option to change their profile photo for a personalized experience.
- Change Password:
 - For added security, users can change their account password anytime directly from this page.

This page provides a simple, elegant interface that ensures users can easily control and secure their information while enjoying a friendly visual design that aligns with the plant store's natural theme.

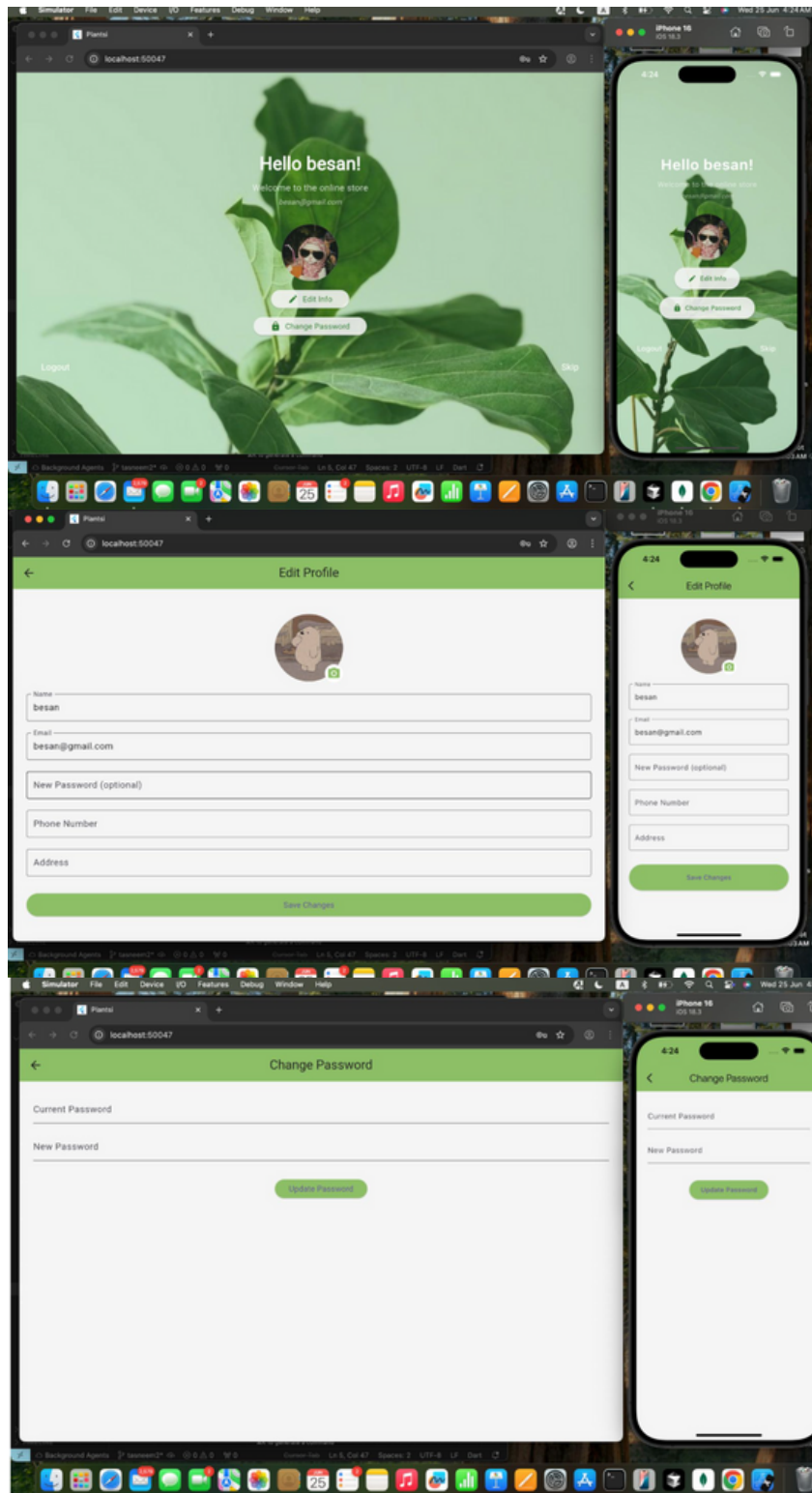


Figure 4.11: Profile Page.

• Shop Owner Dashboard

In this page, all plants in the shop are displayed as cards, showing key details like:

- Plant name
- Price
- Available quantity
- Quality
- Type (Indoor / Outdoor / Decorative)

The shop owner can edit any plant using the ✎ icon or delete it using the 🗑 icon.

• Recent Orders

Below, all customer orders are listed with details including:

- Total amount
- Payment method (cash, card, PayPal)
- Date and time
- Customer info (email, phone, address)
- Ordered items with quantity and price

Orders can also be deleted using the 🗑 icon to keep the dashboard clean and up to date.

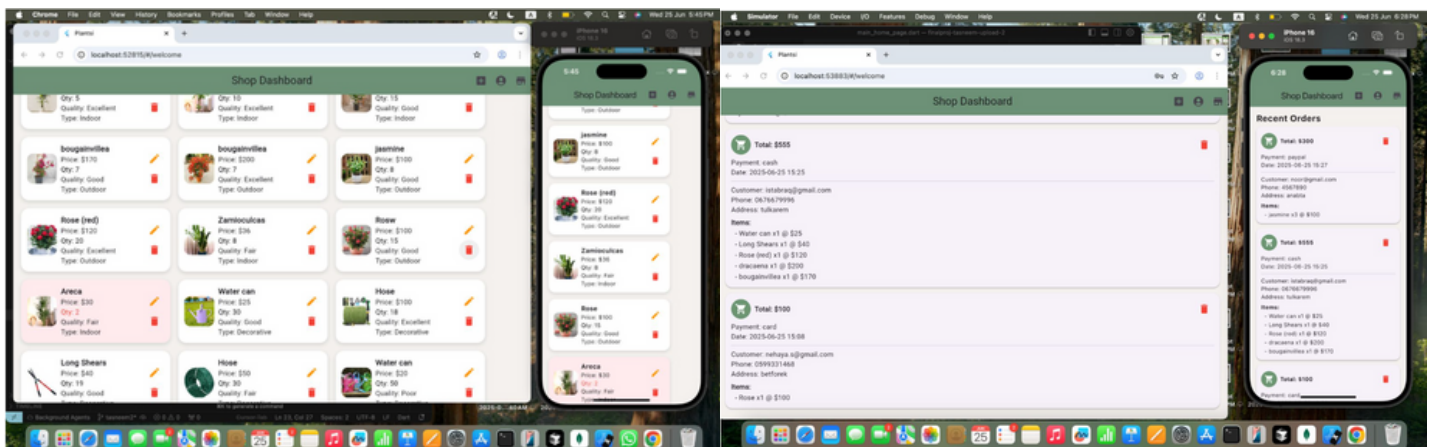


Figure 4.12: shop owner dashboard.

• Add new plant

This page shows the "Add New Plant" interface, where the shop owner can enter the plant name, price, quantity, quality, type, and upload an image. After that, clicking Save Plant will add it to the store. The interface works on both web and mobile.

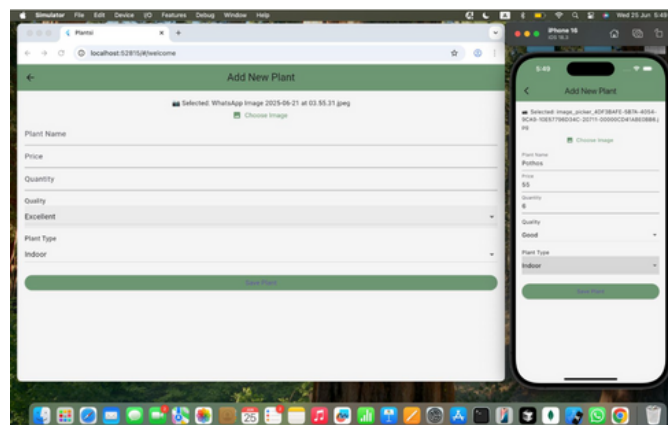


Figure 4.13: Add new plant.

- Edit shop info

This page displays the Edit Shop Info interface, where the shop owner can update the shop's details such as:

- Shop Name
- Shop Address
- Phone Number
- Shop Description
- Specialties (Indoor, Outdoor, Decorative plants)

At the bottom, a list of the shops added by the user is shown, with options to edit or delete each shop using the appropriate icons.

The interface is fully responsive and works smoothly on both web and mobile views.

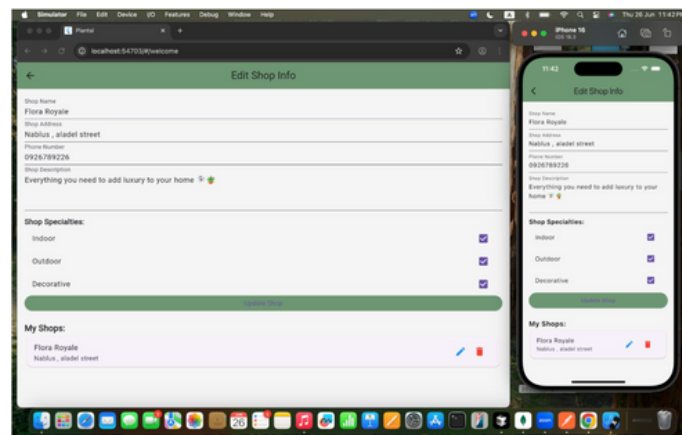


Figure 4.14: Edit shop info.

- user main page

- Search bar: At the top, users can search for a specific shop by name.
- Shop display: Shops are shown as cards, each containing:
 - The shop name
 - A short description
- Responsive layout: The design appears clear and organized on both web and mobile views.
- Quick access icons: At the top, there are shortcut icons for navigation such as home, favorites, orders, profile, etc.

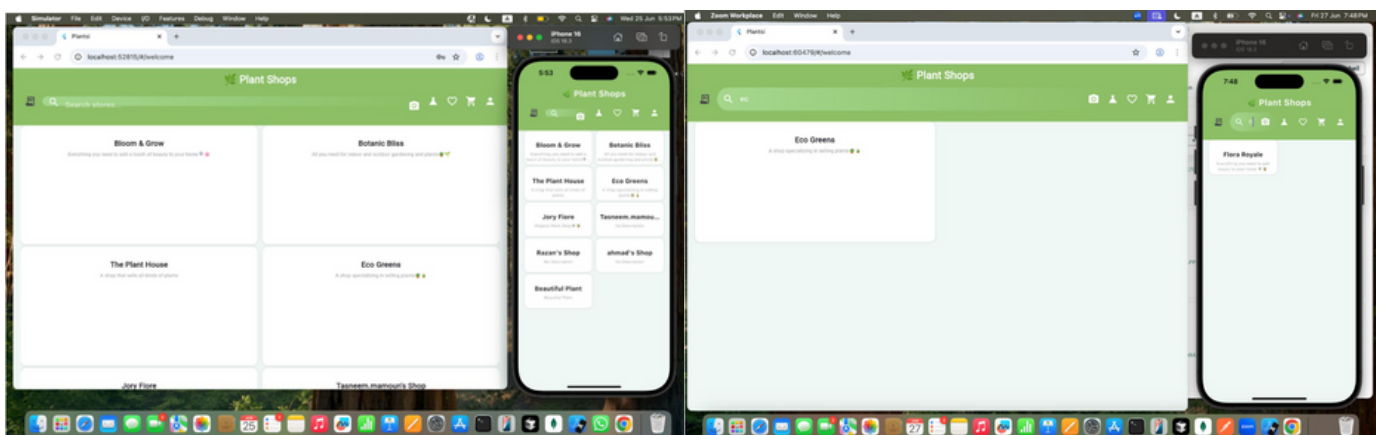


Figure 4.15: user main page.

- View plants and decorative

When entering any shop, all its plants are displayed in a neat interface showing the image, name, and price, with filtering options (Indoor, Outdoor, Decorative) and a button to add items directly to the cart for purchase.

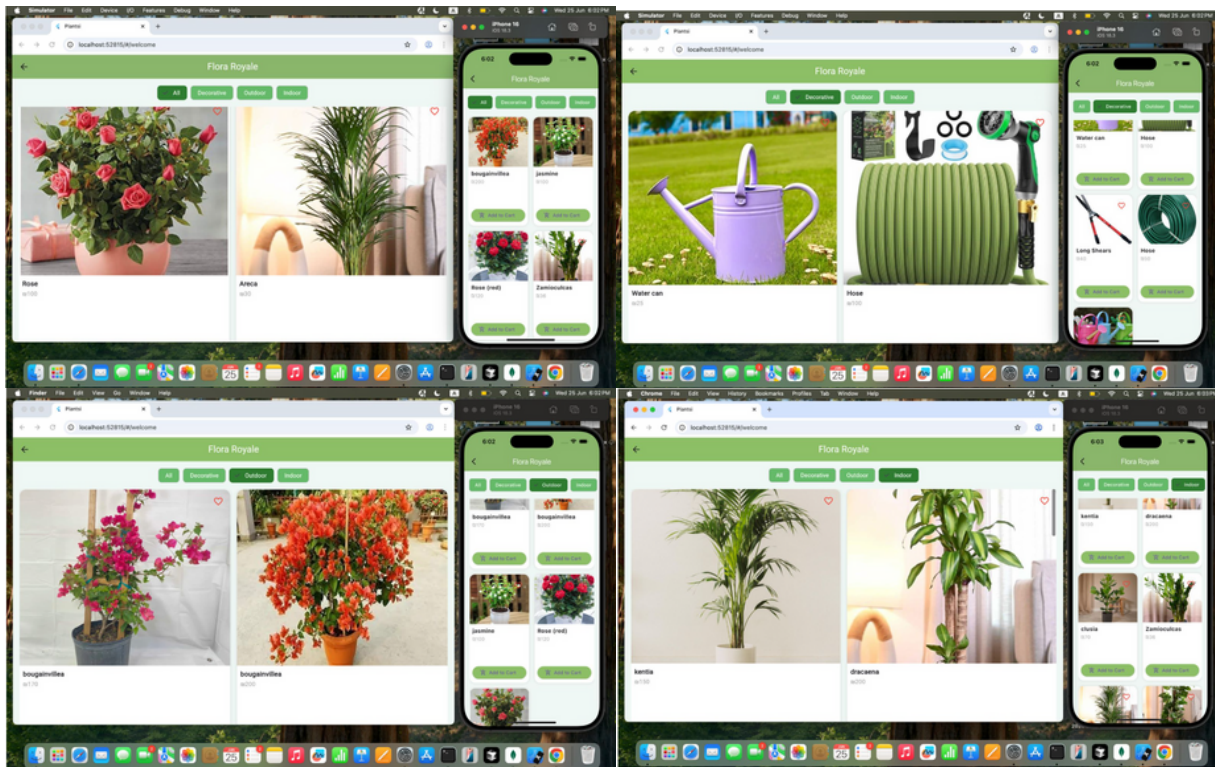


Figure 4.16: View plants and decorative.

- View plants info

When a plant is selected, it is displayed on a separate page with a clear image, the plant's name at the top, and its price at the bottom. There are also buttons to add the plant to the cart or view more details on Wikipedia.

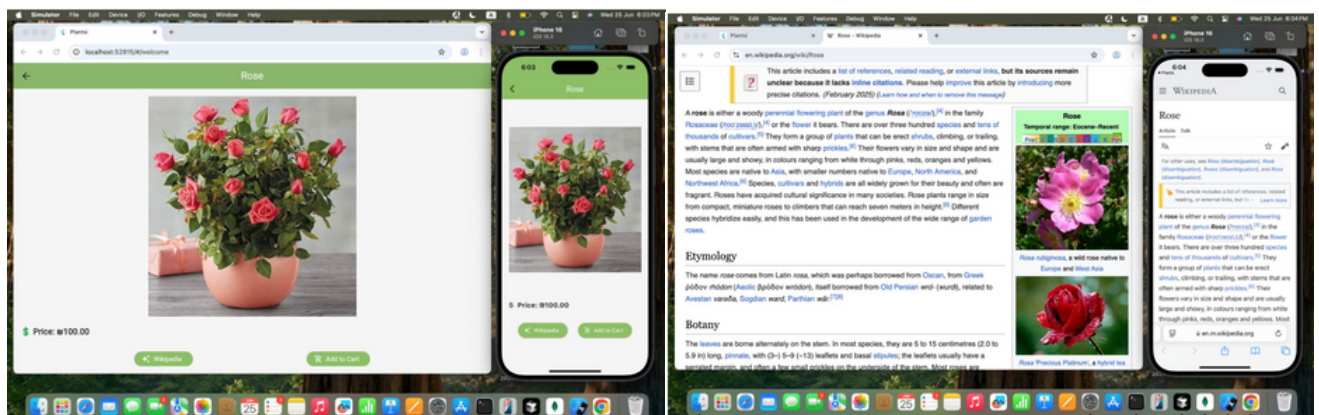


Figure 4.17: View plants info.

- Favorites

In this “Favorites” page, the plants that the user has added to their personal list are displayed. Each item shows the plant’s image, name, and price, with the option to remove it using the 🗑 icon.

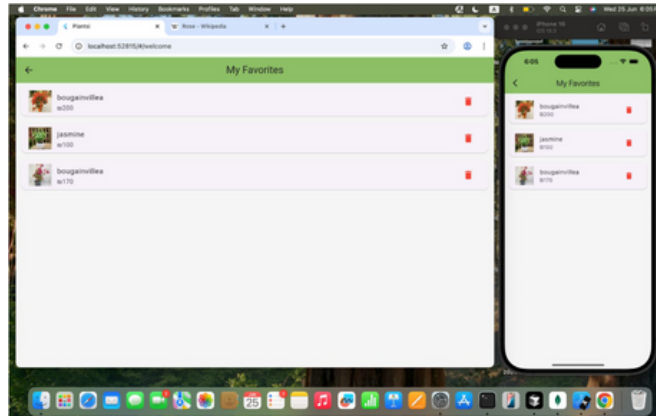


Figure 4.18: Favorites.

- cart

These images show the shopping cart and checkout screens in the app. The user can view added items with their prices, adjust quantities, or remove items. A coupon code can also be applied. Then, the user proceeds to checkout by selecting a payment method (Card, PayPal, or Cash), entering card details and the shipping address to complete the order easily.

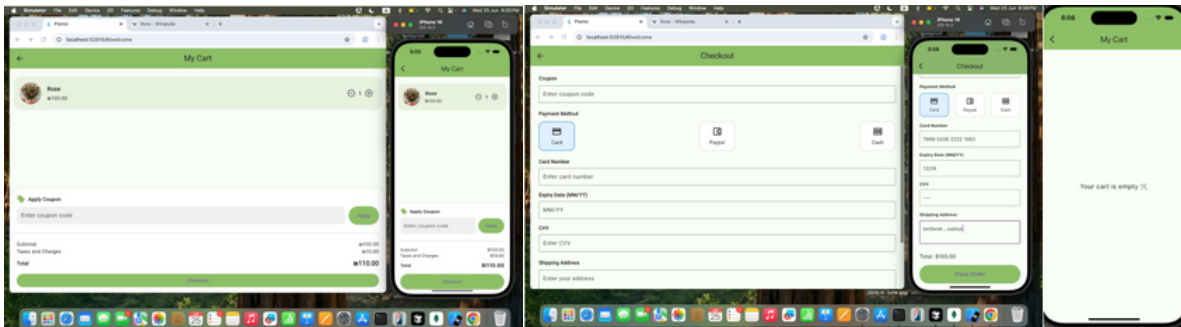


Figure 4.19:cart.

- My orders

This screen shows the "My Orders" page for the user, where they can view details of their previous order such as: order number, current status (Pending), total amount, and order date. The interface is available on both web and mobile.

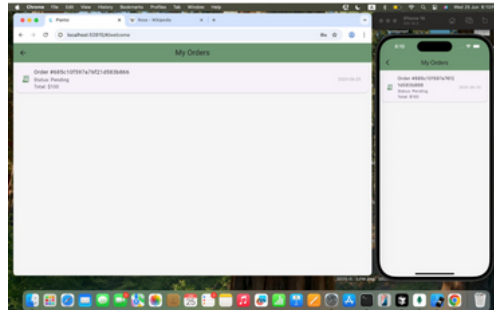


Figure 4.20: my order.

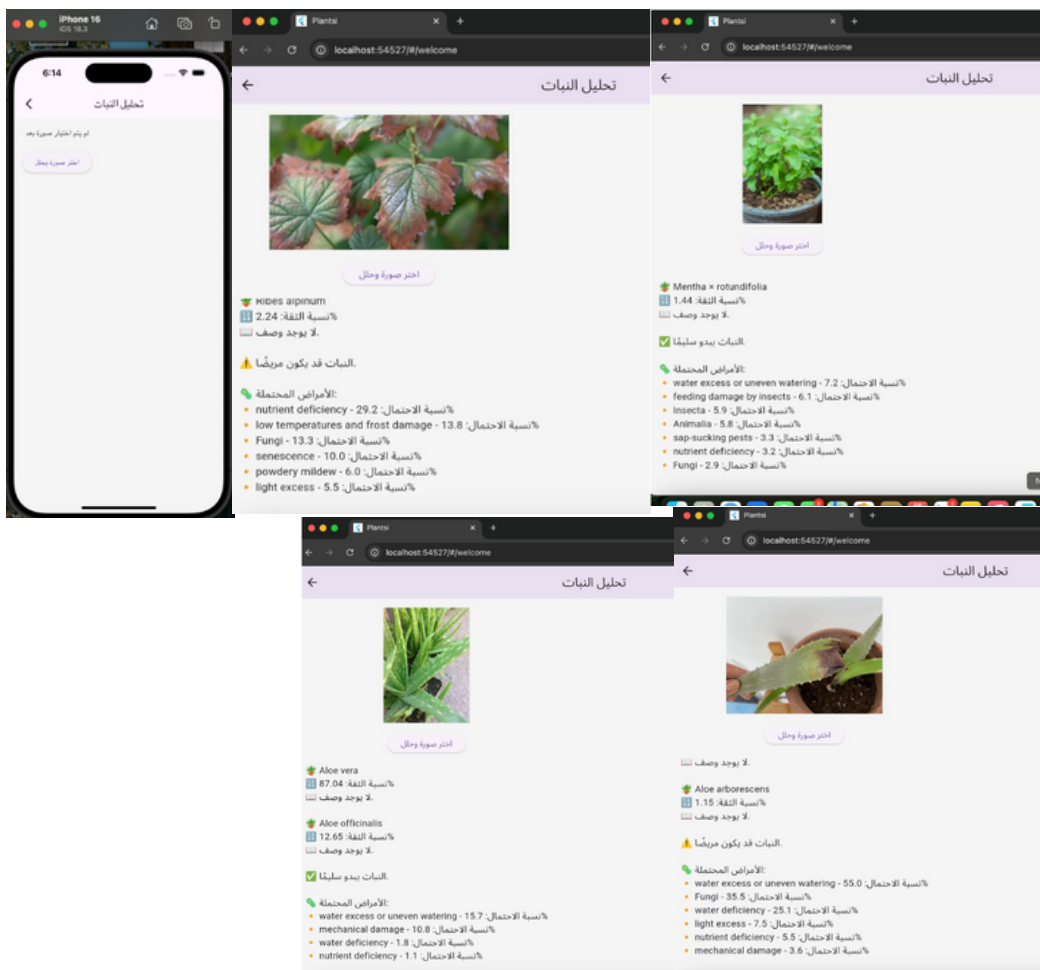


Figure 4.21: Ai Analysis.

- **Ai Analysiz**

The Plant Analysis feature allows users to upload a photo of their plant to be analyzed using AI. After uploading, the app displays the plant's name with a confidence percentage. If the plant appears unhealthy, the app shows possible diseases along with the likelihood of each one, such as over/underwatering, fungi, mechanical damage, or nutrient deficiency. If the plant is healthy, a message like “The plant looks healthy ✅” is shown.

- **Sales Trend**

The image shows a Shop Dashboard that includes a Sales Trend chart displaying the monthly sales performance. The chart illustrates the number of orders per day throughout the month, helping the shop owner analyze sales activity and identify peak or low-demand periods.

✅ **Benefit:** This kind of analysis helps in making better decisions, such as boosting offers on slow days or focusing on popular products.

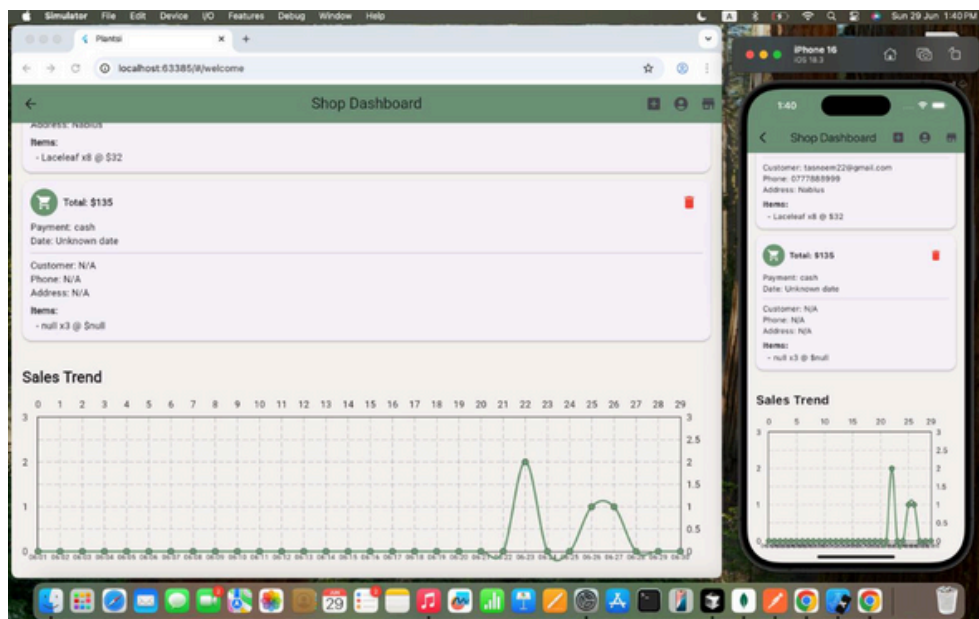


Figure 4.22:Sales Trend.

The system works as follow

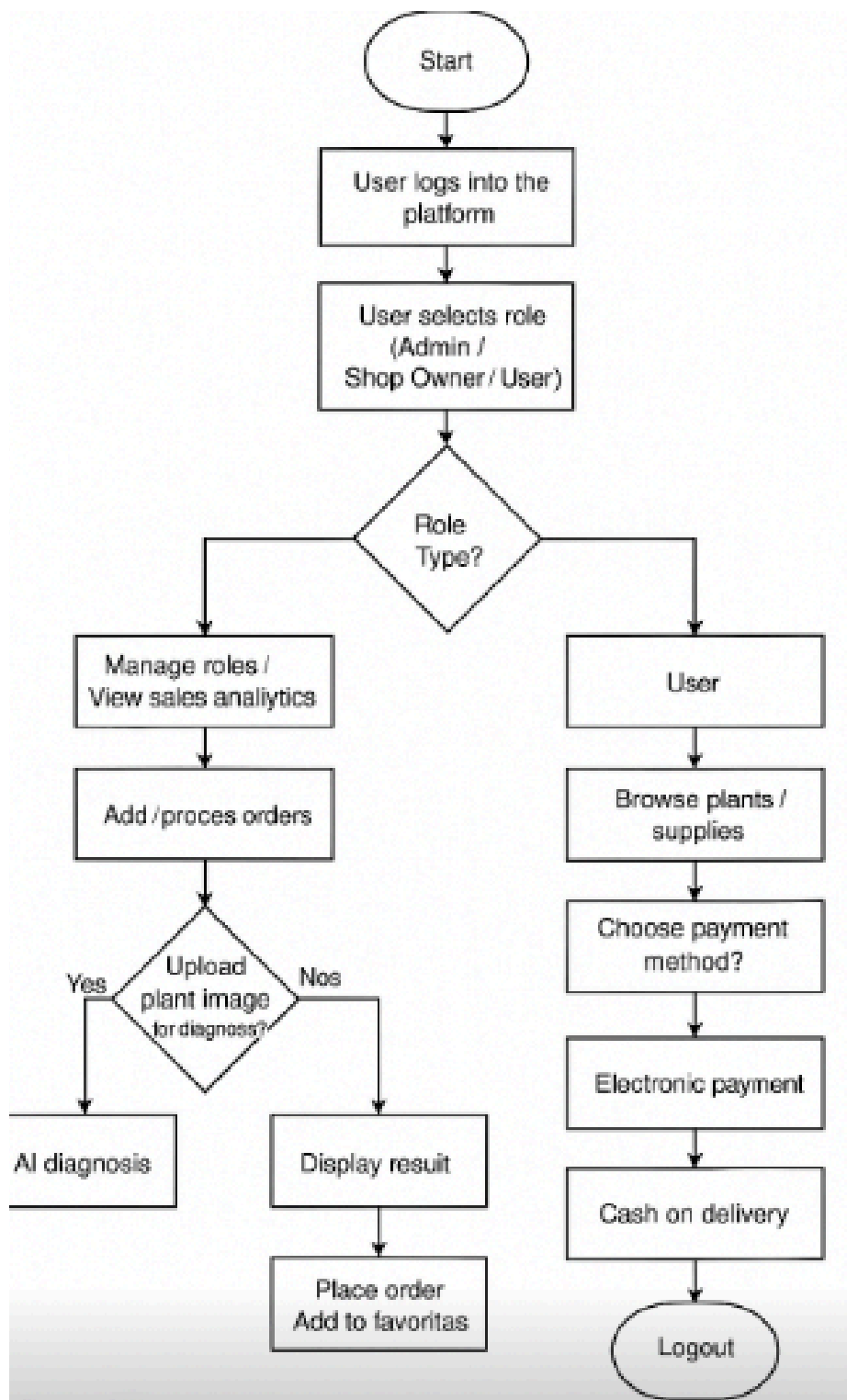


Figure 4.23: Flow Chart of the system.

Chapter 5: Results and Analysis

5.1 Final Application:

The final Plantsia application successfully integrates all the required functionalities, providing an intuitive and efficient user experience for admins, shop owners, and users. The system allows shop owners to manage their shops and products, users to browse, order, and favorite plants, and admins to oversee the platform's performance and manage roles. Additionally, the AI-powered plant disease diagnosis feature provides users with instant feedback and recommendations.

Key features of the final application include:

- **Shop and Product Management:** Shop owners can add, edit, and delete plants and agricultural supplies, and manage orders efficiently.
- **AI-Based Plant Disease Diagnosis:** Users can upload images of their plants to detect diseases, identify causes, and receive treatment suggestions.
- **User Profiles & Statistics:** Each user has a profile displaying their order history, favorites, and interactions.
- **Sales Analytics:** Admins can view sales ratios and insights to monitor platform performance.
- **Secure Role Management:** Admins can add, edit, or delete user roles and manage shop owners.
- **The application is designed with scalability in mind,** ensuring smooth performance as the user base and data grow. Feedback from initial users confirms that the platform meets its intended objectives effectively.

Project Outcomes

The project has achieved its primary goals by successfully implementing a modern platform that enhances plant commerce through smart features and user-friendly design.

The following outcomes have been observed:

1. **Improved Plant Shopping Experience:** The platform provides a flexible and interactive experience for plant lovers and shop owners.
2. **Efficient Shop Management:** Shop owners can easily handle product listings and orders, improving operational efficiency.
3. **Smart Features:** The integration of AI diagnosis and sales analytics supports informed decision-making and personalized service.
4. **Secure and Scalable Design:** The system ensures data security and is built to accommodate future growth and enhancements.
5. **Foundation for Future Development:** The modular structure of the application allows for future additions, such as AI-based plant recommendations and integration with payment gateways.

In conclusion, Plantsia meets its objectives successfully and lays a strong foundation for further enhancements. Future improvements could include personalized plant care guides, advanced search filters, and expanded agricultural supply categories to reach a wider audience.

Challenges

- **Challenges with Image Accuracy:** Environmental factors such as lighting conditions or image quality sometimes affect the accuracy of the AI-based plant disease diagnosis, leading to potential misclassification or missed detections.
- **Variations in Plant and Supply Data:** Differences in how shop owners enter plant details or agricultural supply information can lead to inconsistencies, requiring data validation and standardization efforts to ensure quality and uniformity across the platform.
- **Challenges in System Performance:** As the number of users, shops, and orders increases, maintaining system responsiveness and performance can be challenging. The system may require optimization or upgrades to handle higher loads efficiently.
- **Challenges in Remote Access Reliability:** Occasional internet connectivity issues or server-side delays can affect the reliability of remote operations, such as shop management and user interactions through the app.

Chapter 6: Conclusions and Future Work

6.1 Conclusions

In this project, an integrated platform for plant and agricultural supply commerce was successfully designed and implemented, streamlining processes from shop management to order handling and AI-based plant disease diagnosis. The system demonstrated high efficiency in simplifying operations, enhancing the shopping experience, and supporting both shop owners and users compared to traditional plant commerce methods. Features like flexible product management, real-time sales analytics, and smart plant health detection added significant value, making the platform adaptable to a wide range of user needs.

While most objectives were successfully achieved, such as improving efficiency and ensuring data consistency, some minor challenges were observed, including occasional variations in image analysis accuracy and performance fluctuations under high load. These challenges highlight areas for future optimization and enhancements.

6.2 Future Work

Based on the results and challenges identified during the system's implementation, several areas for improvement and expansion have been identified for future development:

1. **Enhanced AI Diagnosis:** Improve the accuracy of plant disease detection by training the model on larger and more diverse datasets, and incorporating additional plant species.
2. **Personalized Recommendations:** Implement AI-driven recommendations for plants and agricultural supplies based on user preferences and purchase history.
3. **Mobile App Enhancements:** Expand mobile application features to include push notifications, order tracking, and plant care reminders.
4. **Multi-Language Support:** Develop multi-language functionality to serve a broader user base and improve accessibility.
5. **Performance Optimization:** Further optimize system performance to ensure scalability and responsiveness as the platform grows.

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